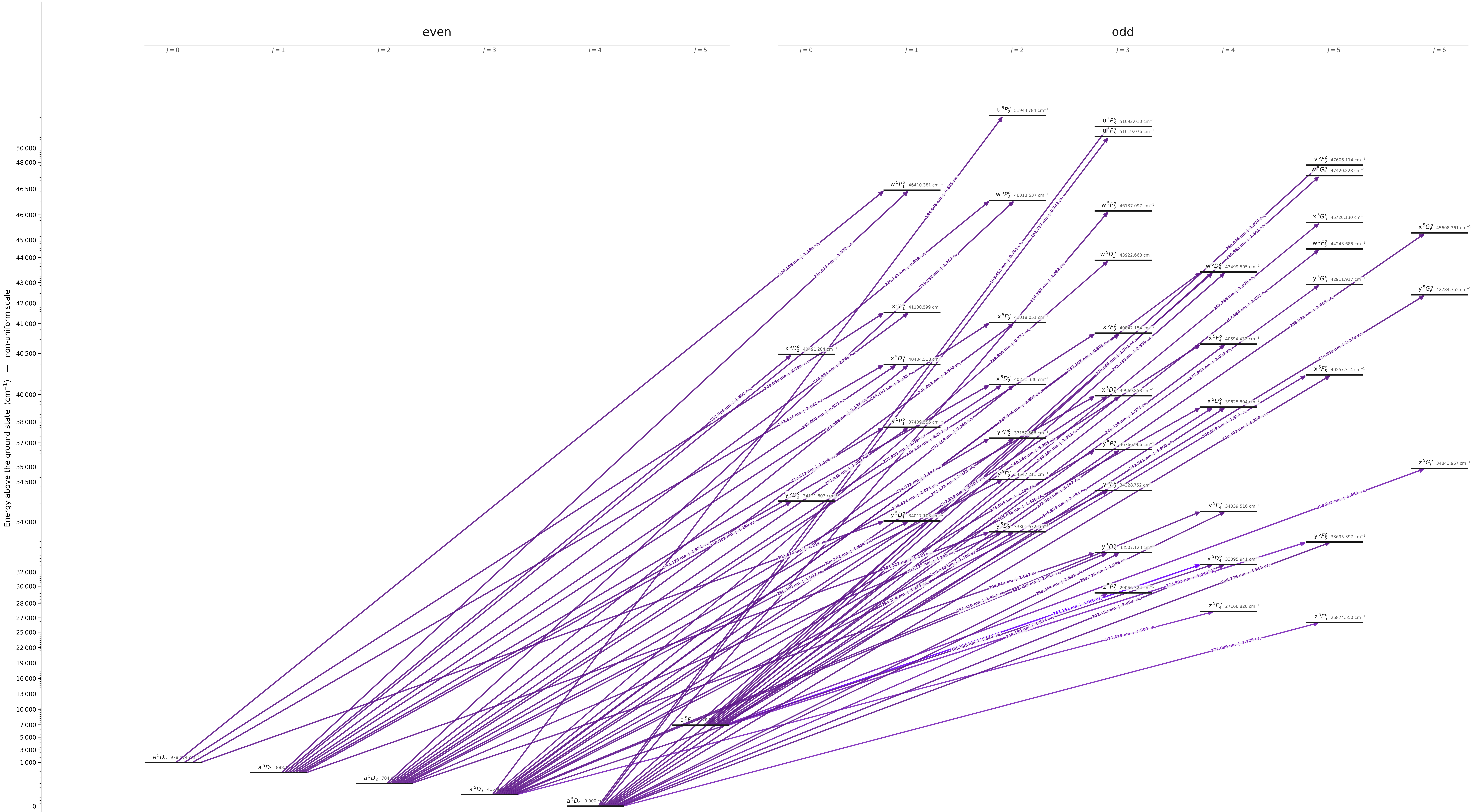
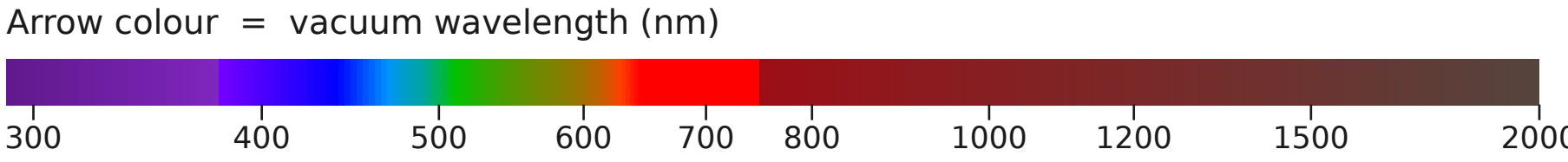


Iron

Fe (Fe I). Strongest classified transitions below 2 μm (NIST): 47 levels, 70 lines.

Fe⁺ ionisation limit 63737.70 cm⁻¹ = 7.90247 eV (156.893 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(U||e_r||J')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er J'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁵ D ₄ - w ⁵ P ^o ₂	216.7453	216.6773	1383.1554	3.082	2.700e+08	-	-	
a ⁵ D ₂ - w ⁵ P ^o ₂	219.2524	219.1839	1367.3393	1.767	1.200e+08	-	-	
a ⁵ D ₃ - x ⁵ F ^o ₁	247.3642	247.2895	1211.9476	2.607	1.300e+08	-	-	
a ⁵ D ₂ - x ⁵ F ^o ₂	248.0525	247.9776	1208.5846	2.560	1.740e+08	-	-	
a ⁵ D ₄ - x ⁵ F ^o ₂	248.4021	248.3271	1206.8839	6.320	4.800e+08	-	-	
a ⁵ D ₁ - x ⁵ F ^o ₁	248.4937	248.4187	1206.4388	2.266	2.260e+08	-	-	
a ⁵ D ₃ - x ⁵ F ^o ₁	248.8893	248.8142	1204.5211	5.363	4.200e+08	-	-	
a ⁵ D ₀ - x ⁵ F ^o ₁	249.0503	248.9752	1203.7424	2.299	2.310e+08	-	-	
a ⁵ D ₂ - x ⁵ F ^o ₃	249.1396	249.0644	1203.3114	4.287	3.440e+08	-	-	
a ⁵ D ₁ - x ⁵ F ^o ₃	249.1906	249.1155	1203.0647	3.333	2.910e+08	-	-	
a ⁵ D ₃ - x ⁵ D ^o ₂	251.1591	251.0835	1193.6357	2.246	1.290e+08	-	-	
a ⁵ D ₂ - x ⁵ D ^o ₁	251.8859	251.8101	1190.1914	2.137	1.930e+08	-	-	
a ⁵ D ₄ - x ⁵ D ^o ₁	252.3608	252.2849	1187.9517	3.900	2.130e+08	-	-	
a ⁵ D ₁ - x ⁵ D ^o ₀	252.5052	252.4292	1187.2726	1.602	3.230e+08	-	-	
a ⁵ D ₃ - x ⁵ D ^o ₂	252.8194	252.7434	1185.7967	3.283	1.930e+08	-	-	
a ⁵ D ₄ - y ⁵ P ^o ₂	271.9833	271.9027	1102.2459	3.142	1.420e+08	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).